Lab: Thinking Fast and Slow -- Your Brain's Two Speeds

Objective

> **Learning Goal:** Explore how your brain uses fast automatic thinking and slow deliberate thinking to update beliefs and make decisions.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- Pencil and paper
- Timer or stopwatch (phone works)
- A partner or small group
- Lab journal or notebook

Part 1: Fast Thinking Challenge (10 min)

Your brain has a "fast mode" that makes snap judgments without you even trying. Let's test it.

Speed Round: Answer each question as FAST as you can. Write your first answer, then take time to think carefully and write a second answer.

Question	Fast Answer (first instinct)	Slow Answer (after thinking)	Were they different?
A bat and ball cost $1.10 total. The bat costs $1 more than the ball. How much is the ball?			
If 5 machines make 5 widgets in 5 minutes, how long for 100 machines to make 100 widgets?			
In a lake, lily pads double in area every day. It takes 48 days to cover the whole lake. How long to cover half?			

Write your response here

Part 2: Belief Updating Game (15 min)

Active Inference says your brain is constantly updating its beliefs based on new evidence. Let's see this in action.

Mystery Box Game: Your teacher (or a group member) hides an object in a box. You get to ask YES/NO questions to figure out what it is. Track how your belief changes with each clue.

Clue #	Question Asked	Answer	My Top Guess Now	Confidence (1-10)
0 (before any clues)	--	--		
1				
2				
3				
4				
5				

Write your response here

Part 3: Cognitive Bias Spotter (10 min)

Your brain takes shortcuts that sometimes lead to mistakes. These are called **cognitive biases**. See if you can spot them.

For each scenario, identify whether the person is using fast thinking, slow thinking, or a biased shortcut:

Scenario	Fast, Slow, or Biased?	Explain
You always pick the same lunch because you know you like it		
You study for a test by re-reading your notes three times and making flashcards		
You assume a new student is unfriendly because they are quiet		
You solve a math problem step by step on paper		
You believe a rumor because everyone else seems to believe it		

Write your response here

Part 4: Design Your Decision (5 min)

Think of a real decision you need to make soon (what to do this weekend, how to study for a test, how to handle a conflict with a friend).

Step	Your Decision
What is the decision?	
What does your fast brain say?	
What new information could you gather?	
After slow thinking, what would you decide?	
How confident are you (1-10)?	

Write your response here

Reflection Table

Question	Your Answer
What is the difference between fast and slow thinking?	
What is belief updating?	
Why does your brain prefer fast thinking most of the time?	
When is fast thinking helpful? When is it risky?	
How does Active Inference explain cognition?	

Write your response here